TANZANIA HEADS OF ISLAMIC SCHOOL COUNCIL

FORM IV INTER ISLAMIC MOCK EXAMINATION 2024

BIOLOGY 1

MARKING SCHEME

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	A	B	C	D	B	B	A	D	C

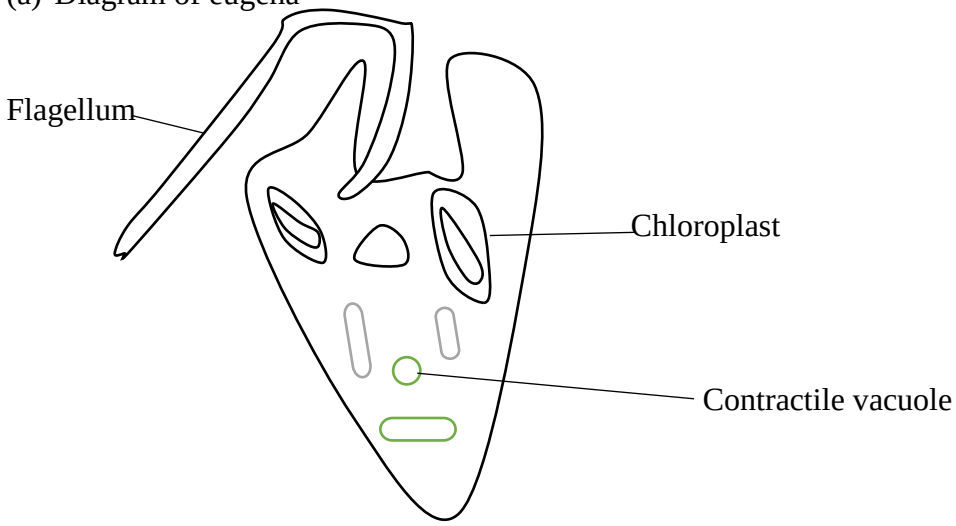
01 mark @ = 10 Marks

2.

COLUMN A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
COLUMN B	C	D	B	E	A	F

01 mark @ = 06 marks

3. (a) Diagram of eugena



Caption = 01

Diagram = 03

Labelling = 2 01 = 03

07 MARKS

4. (a) Mutualism:

This is the relationship in which two organisms benefit each other. Example the rhizobium bacteria in the root modules of legumes convert nitrogen into nitrates of sue by plants. The bacteria get protection nutrients from the plants. **1.5**

Commensalism:

This is interaction that is beneficial to one organism and is neutral to the other organism. Example when bird build a nest in a hole, in a tree. **1.5**

Parasitism:

In this association one organism benefits while the other is harmed. Example plasmodium that causes malaria in human beings. **1.5**

(b) Feeding relationship is food chain:

Grass → Grasshopper → Toad → Snake → Bacteria 0**2**

(c)

Trophic level	Organism
Producer	Grass
Primary consumer	Grasshopper
Secondary consumer	Toad
Tertiary consumer	Snake
Decomposer	Bacteria

@ = 02.5 marks

5. (a) Osmosis is the movement of water molecules form the region of high concentration of water to the region of low concentration of water through the semi-permeable membrane. **02**

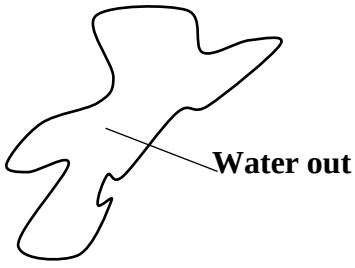
(b) When red blood cells are put in a hypotonic solution, they absorb water, causing the cell volume to increase, Excessive amount of water cause HAEMOLYSIS (bursting). **02**

When red blood cells are put in a hypertonic solution, they lose water, leading to riveling of the cell. This is referred to as GRENATIN. **02**

Hypotonic solution **Hypertonic solution**



01 ½ **1 ½**



Haemolysis **Crenation**

6. (a) Functions of skeleton.

(i) Support:
The skeleton provides rigid frame work which supports soft parts of the body. It provides attachment for muscles and body organ?

(ii) Locomotion:
The skeleton enables the organism to move from one place to another.

(iii) Protection:
Skeleton protects delicate internal organs e.g. brain, heart.

(iv) Formation of blood cells:
Red blood cells and white blood cells are made in the bone marrow.

(v) Skeleton store minerals such as calcium and phosphorus.

(vi) Shape: Skeleton animals definite shape.

@ 01 = 05

(Any five)

(b)

	Skeletal muscle	Cardiac Muscle
(i)	Voluntary muscles	Involuntary muscles
(ii)	It get fatigue	It does not get fatigue
(iii)	Its has no pacemaker	It has pacemaker
(iv)	Found on skeleton	Found in heart
(v)	Few mitochondrion	Numerous mitochondrion

(Any two)

@ 02 = 04 marks

7. (a) If heterozygous man cross with heterozygous woman.

Parental phenotype Carrier x Carrier **01**

Parental genotype Hb^AHbs x Hb^AHbs

Meiosis

Gametes Hb^A Hb^s HB^A Hbs **01**

Fertilization

F1 - genotype Hb^AHb^A Hb^AHb^s Hb^AHb^s Hb^sHb^s **o1**

F1-Phenotype Nominal Carrier Carrier Sickle cell disease **01**

Therefore the possible phenotypes of the offsprings are:

1 - Normal = 25%

2 - Carrier = 50%

1 - Sickle cell disease = 25% **@ 01 = 03**

(b) (i) Severely anaemic $= \frac{1}{4} \times 100\% = 25\%$ **01**

(Hb^sHb^s)

(ii) Mild anaemic $= \frac{2}{4} \times 100\% = 50\%$ **01**

(Hb^AHb^s)

8. (a) (i) A – Dendrites

B - Cell body **@ ½ = 02**

C – Axon

D – Myetin sheath

(ii) Motor neutone carries impulses from the central nervous system to the effector organ causing reaction to take place.

(iii) Sensory neurone and relay neurone. **01**

- Sensory neurone carries information form the sense organs to the central nervous system for interpretation. **01**
- Relay neurone interlinks sensory and motor neurons in the central nerous system. **01**

(b)

Nervous Communication	**Endocrine communication**
Communication is through electrical signals	Communication is through hormones
The response is fast	The response is slow
The response is short lasting	The response is long lasting
Electrical signals are transported through neurons	Hormones are transported by bloods

Any two @ 1 ½ = 03

9. (a) Elderly people require:

- Vitamins, minerals and roughage and {fruits vegetables, meat, fish, eggs, etc}
- Proteins

(b) Expectant and pregnant mother

- Vitamins
- Minerals

} fruits, vegetables, eggs, fish, meat

- Proteins
- Roughage

(c) Mammal works

- Carbohydrates
- Proteins
- Minerals
- Lipids
- Vitamins

maize, flour, meat, fish, vegetables, grains, fruits

(c) Sick persons

- Proteins
- Vitamins
- Minerals
- Carbohydrates

Eggs, fish, meat, fruits, vegetables, maize, etc

Introduction = 1 ½

Main body @ 2 ½ = 12 ½

Conclusion = 01

TOTAL 15 Marks

10. Introduction: Meaning of water cycle. The water cycle involves the following seven stages.

(i) Water from ground and run off flow into streams and rivers.
(ii) The streams and rivers flow into lakes and oceans.
(iii) Water from water bodies such as oceans, lakes, ponds and rivers evaporate into the atmosphere. Similarly, water form plants evaporates to the atmosphere through transpiration to form water vapour.
(iv) Water evaporating from water bodies is discharged to the atmosphere and condenses to form clouds.
(v) Wind causes clouds to move in the atmosphere above the ocean and move further inland.
(vi) As condensation continues the air in the atmosphere becomes saturated to the next that it can not hold more moisture. The clouds becomes target and too heavy to be moved by wind.
(vii) Therefore, clouds start to precipitate and fall in drops to the ground and water bodies as rain or snow. Some water is absorbed by plants and some flow into water bodies and cycle begin again.

Conclusion: Any relevant points.

Introduction 0 ½
Main body @ 02 = 14
Conclusion = 0 ½

2

TOTAL 15

11. (a) The menstrual cycle: Is the monthly discharge of blood mucus and epithelial cells from the uterus through the vagina.

(b) The menstrual cycle is regulated by the following hormones.

(i) Gonadotrophic hormone: It stimulates the anterior pituitary gland to secrets follicle stimulating hormone (FSH).

(ii) Follicle stimulating hormone (FSH): Stimulate the development of several follicles in the ovary livings.

(iii) Oestrogen hormone: It stimulate repair and development of the endometrium in the uterus. This prepares for the possibility of pregnancy.

(iv) Lutenizing hormone (L.H): It influences ovulation in the ovary.

@ 02 = 08

(c) Meal rich in seal food like soyabeans, beans, pea and sorghum, these stimulate the secretion of blood suice contain iron minerals. **02**

Introduction = 02

Conclusion = 01

TOTAL = 15

Wabillah Tawfiq